

Multiple Choice (10 marks)

1. The lunar maria are:
 - (a) ancient heavily cratered highlands
 - (b) dark lavas inside volcanic calderas
 - (c) dark lavas filling older impact basins
 - (d) the bright regions on the Moon
2. Why are Cepheid stars relevant for astronomers?
 - (a) To measure interstellar mass.
 - (b) To measure galactic distances.
 - (c) To measure galactic energy-density.
 - (d) To measure interstellar density.
3. The sky is blue because
 - (a) the Sun mainly emits blue light.
 - (b) the atmosphere absorbs mostly blue light.
 - (c) molecules scatter red light more effectively than blue light.
 - (d) molecules scatter blue light more effectively than red light.
4. The terrestrial planet cores contain mostly metal because
 - (a) the entire planets are made mostly of metal.
 - (b) metals condensed first in the solar nebula and the rocks then accreted around them.
 - (c) metals sank to the center during a time when the interiors were molten throughout.
 - (d) radioactivity created metals in the core from the decay of uranium.
5. Which one of these constellations is not located along the Milky Way in the sky?
 - (a) Perseus
 - (b) Cygnus
 - (c) Scorpius
 - (d) Leo

True / False (10 marks)

Answer True or False

6. True or False: Knowing both the actual brightness and apparent brightness of an object allows you to determine its composition.
7. True or False: The ratio of solar radiation flux on Mercury's surface at perihelion (0.304 AU) compared to aphelion (0.456 AU) is 9:4.
8. True or False: Differentiation occurs solely due to spontaneous emission from radioactive atoms in a planetary interior.
9. True or False: The visible part of the electromagnetic spectrum is typically defined as spanning from 240 to 680 nanometers.
10. True or False: The part closer to the axis of a solid disk rotates with a longer period than the outer regions of the disk.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the Laniakea supercluster in the context of the Milky Way's position within the universe. Analyze its impact on our understanding of cosmic structure.
12. Discuss the definition of a parsec in astronomy, including its significance and the method by which it is determined in relation to astronomical units and angles.

End of Paper